

Machine Vision System for Autonomous Agricultural Vehicle

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Abstract—The problem of the vehicle motion control for automated driving is considered. The robotic chassis is designed for agriculture and should work in the absence of roads. The general structural diagram of the chassis' machine vision system is given. A structural obstacle separation method for constructing obstacle maps on the ground is proposed. The method uses lidars to detect obstacles. The implementation of the method is based on the assumption that the terrain within the area no significant differences in elevation (gullies, dips) and water obstacles. For solving the detecting obstacle problem in these conditions, it is enough to detect points which height exceeds a certain threshold, and to identify the relationship between these points to assess the obstacle size. The clustering of points in each layer by the Euclidean distance is performed. Then, the coordinates of the cluster centers are recalculated into the global coordinate system, which allows transferring obstacles to an area map, taking into account the dimensions that determine the degree of the detected obstacle danger. An algorithm for implementing the proposed method is described. The report also provides information on the composition of the software for the robotic chassis machine vision. Simulation results of the proposed method of the allocation of obstacles are presented. The method has low computational complexity, which reduces the requirements for the robotic chassis on-board computer.

Keywords—self-driving vehicle, agricultural robots, machine vision, obstacle detection

I. INTRODUCTION

Robotic transport systems for various purposes perform many production functions. Nowadays, the use of robots in agriculture is actively developing. This trend leads to the creation of robotic vehicles: combines, trucks, tractors, designed for a wide range of technological operations [1– 6]. The complexity of the management tasks of autonomous agricultural vehicles lies in the variety of external conditions in the fields and the difficulty of identifying natural objects. Several fundamental works are known devoted to various aspects of the agricultural robotic systems design and operation [7–9].

The presented report discusses the vision system - one of the main systems for control the motion of a robotic chassis (RS). The RS is included in the integrated robotic system of agricultural vehicles and is a modified model of KAMAZ 65119 [10].

The autonomous control algorithm implies the independent motion of the RS along the route previously developed and approved by the upper level of the robotic system, harmonized with the routes of other RS. Research in the field of creating motion control systems for RS is carried out in two directions [11–15]:

- Development of methods, algorithms and technical means for driving a single agricultural vehicle;
- Development of a multi-level management system, including a group of RCs, functioning simultaneously and independently from each other.

The problem of the RS unmanned control is solved by creating a machine vision system. Existing robot vision systems are based on the most common tools for navigating in space, which are lidars, radars, video cameras [16, 17]. The measurement data are used to build maps of the area [18–20]. However, the described systems are mainly intended for unmanned vehicles operating in an urban environment. Autonomous vehicles for use in agricultural technological processes must include a reliable obstacle detection system.

Most often, for detecting obstacles, the methods, which are based on the analysis of signals from lidars and radars, are used [21, 22]. At present, the efforts of researchers are aimed at creating intelligent methods for analyzing information about the space surrounding the car. Such methods use fuzzy logic and artificial neural networks [23–26]. Most methods require significant computational resources. It should be noted that existing methods are focused on working with a dense cloud of points at a distance of up to 20-25 meters from the lidar and work at a greater distance may be unstable.

The proposed method for detecting obstacles at a great distance and assessing the degree of danger for a moving RS was developed to overcome the limitations of the above methods.

II. MACHINE VISION ARCHITECTURE OF KAMAZ ROBOTIC CHASSIS

For the motion of autonomous vehicles in the city space or beyond, information is needed on the environment, such as roads, traffic participants, obstacles. The orientation problem in space is the most important in the development of RS. For this purpose, the construction of obstacle maps is carried out based on which the vehicle is navigated in three-dimensional space. Lidars, radars and video cameras are some of the most common

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tools for navigating in space. The many sensor systems is building depth maps by using the reflected signal from scanned surfaces. In this case, the solution to the control problem determines the motion from point A to point B along the optimal path. The choice of the trajectory directly depends both on the vehicle geometrical parameters (dimensions, possible rotation angles), and on external factors - the presence of obstacles, restrictions on movement in predetermined sections of the path.

The structure of the robotic chassis' machine vision system is shown in Fig. 1.

The main system components are:

- sensor subsystem of an unmanned vehicle;
- subsystem of data processing and transmission;
- Operator WorkStation for remote control.

The sensors subsystem includes lidars and cameras for a visual overview of the surrounding space.

There are two types of lidars:

- 2D lidar - for building an array of points in one plane, relative to polar coordinates;
- 3D lidar - for plotting points in volumetric space by rotating the scanning head around the Z-axis, consisting of many lasers mounted one above the other.

The recognition and distance detection module uses obstacle maps. The module uses the developed method of obstacle structural separation. Besides, procedures are carried out to classify obstacles and form an RS motion local route. Data collection from blocks in the RS is carried out via Ethernet. The operator's workstation is at the top level of the robotic vehicle system. Tasks performed in the workstation are related both to

control the position of a single vehicle and to control a "swarm" of robotic chassis during their operation in the field. The operator has the ability to control the RC position on the map. The operator, as appropriate, may also analyze the video stream of the surrounding space images formed by RS sensor subsystem.

The external communication system uses V2X (Vehicle-to-Everything) wireless technology.

III. THE OBSTACLE MAPPING

An obstacle map of the real environment is formed based on data received from the lidar. Lidar is a technology for obtaining and data processing about distant objects using active optical systems that use the phenomena of absorption and scattering of light in optically transparent media.

Lidar advantage over other optical scanners is the ability to work in the absence of illumination since the lidar is itself a source of invisible light.

The most commonly used method is simultaneous localization and map building (SLAM - Simultaneous Localization and Mapping).

The above method used in mobile autonomous means for building a map in an unknown space or for updating a map in a previously known space with simultaneous control of the current location and distance traveled [27, 28]. Popular methods for approximate solving this problem include a particle filter and an advanced Kalman filter. Obstacle detection is one of the main sub-tasks of the SLAM method. In this report, the authors propose a fast and accurate algorithm for the structural separation of obstacles in the field, focused on traffic conditions in agricultural fieldwork.

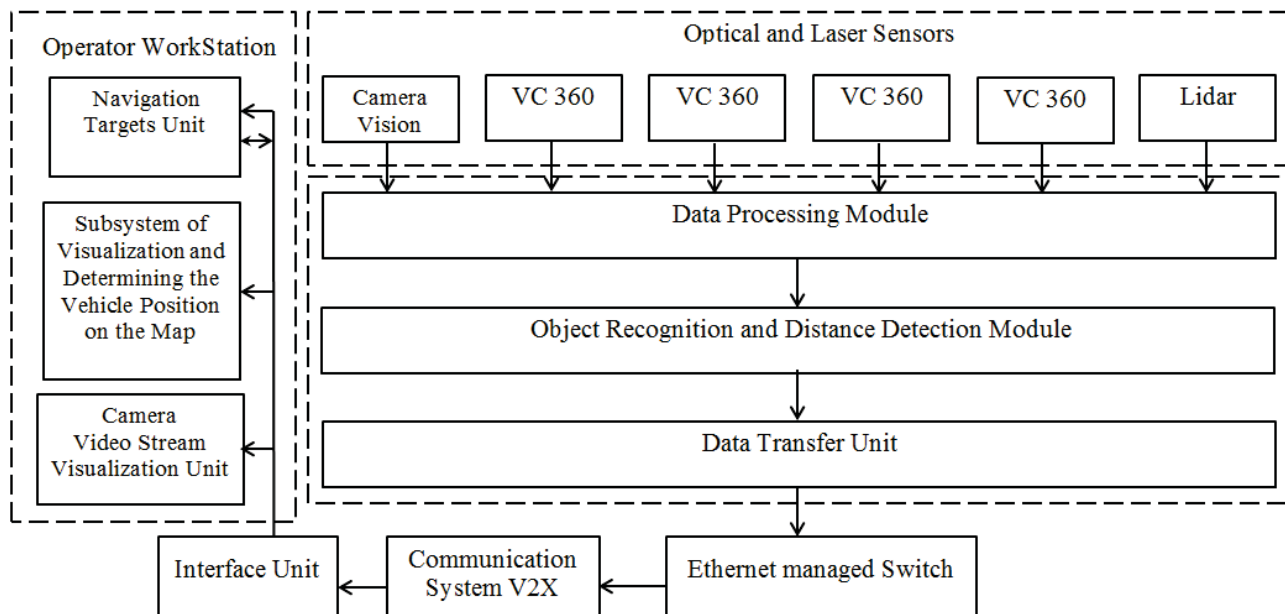


Fig. 1. Structure of the machine vision system for a robotic agricultural chassis

IV. OBSTACLE DETECTION METHOD

The main obstacle detection method is to analyze the data from the sensors, as follows:

- receiving an array (point cloud) - a data set received from sensors represents on distance data to the environment and the angle relative to each scanned area;
- extraction of special points located on faces or planes to identify key objects;
- finding correspondence of points - determination of key objects in the current frame (point cloud), relative to the previous frame;
- motion assessment - determination of own position relative to key points;
- odometry - the placement error assessment of the previous frame relative to the current one, correction in the sensor error.

The essence of the proposed method for detecting obstacles is as follows.

We consider a point cloud obtained from an RS lidar at time. Let the coordinates of the cloud points are given in the right rectangular local coordinate system (LCS) associated with the center of the lidar. While the Oz axis of the LCS is directed upward, and the axis Oy - forward in the direction of travel. It is assumed that the unambiguous rules for the transfer from LCS to the global coordinate system (GCS) are known, associated with the map of the terrain along which the robot moves. The RS geometric proportions and the distance h between the lidar center and the surface point with vehicle location are also set. The assumption is made that there are no significant differences in elevations (ravines, dips) and water barriers on the ground in the area of the robot's operation.

It is enough to detect points whose height exceeds a certain given threshold and to identify the relationships between these points to estimate the size of the obstacles. The point cloud is divided into $n + 2$ horizontal layers using planes parallel to xOy , with a step along the vertical axis equal to Δh , starting from $-h$. The upper and lower layers are discarded, and the remaining layers are numbered from 1 to n .

Also, the features of the point cloud obtained by the lidar are taken into account: the vector between two points lying in the plane passing through the Oz axis on the vertical surface should be collinear to the axis Oz . To simplify the calculations, we will cluster the points in each layer by the Euclidean distance and measure the angle α between the vector connecting the centers of the resulting clusters from different layers and the Oz axis. Let a certain threshold value $\alpha_{\max}$ be given. If $\alpha < \alpha_{\max}$, then the corresponding clusters are considered as connected. The points of these clusters correspond to one object.

Using the clusters interconnected in neighboring layers, we can estimate the height of the object, and by the maximum distance between the points in the clusters - its diameter. The recalculation of the coordinates of the cluster centers in the GCS allows transferring of obstacles to an area map, taking into

account the assessment of the dimensions, which determine the danger degree of the detected obstacle.

The scheme for obtaining a point cloud in a local coordinate system using a multipath lidar and dividing a point cloud into layers is shown in Fig. 2.

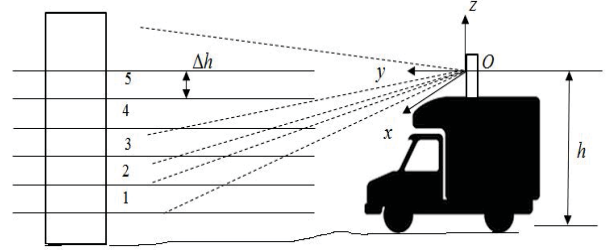


Fig. 2. Point cloud determination model

We denote by M the vector set connecting the centers of connected clusters in different layers. We assume that in each of the selected layers, there will be at least one cluster. Then the method of structural separation of obstacles can be described by the following sequence of actions:

1) set $M = \emptyset$ and select n horizontal layers in the point cloud according to the rule described earlier;

2) points are clustered for each layer, and the number of resulting clusters k_2 , $i = 1, 2, \dots, n$ in the layer определяется is determined;

3) the geometric center CL_2 and the maximum distance between the points of the cluster j , $j = 1, 2, \dots, k_2$ are determined in the layer;

4) for each center of the cluster CL_2 in the layer i , $i = 1, 2, \dots, n - 1$, the nearest center of the cluster CL_{2m} , is located in the layer $i + 1$ and the angle $\alpha_{1,2}$ between the vector $CL_2 - CL_{2m}$ and the positive direction of the Oz axis is calculating; if the inequality $\alpha_{1,2} \leq \alpha_{\max}$ holds, then the vector $CL_2 - CL_{2m}$ belongs into the set M ;

5) vectors belonging to the set M are distinguished with the following property: the endpoint of one vector is the starting point for another and replace them with one vector equal to the sum of the original ones;

6) to determine the position of obstacles on the terrain, the coordinates of the starting points of the vectors are transformed from the set M to the GCS, these points correspond to the centers of obstacles whose height is taken to be equal to the height of the corresponding vector, and the diameter is the largest of the previously estimated dimensions of the object.

The above algorithm was implemented in the prototype of the obstacle detection subsystem. The correct operation of the prototype was tested by adding visualization of detected obstacles. The prototype was verified on test data as separate point clouds. Thus, some dependencies of the performance and obstacle detection range on the values of h , Δh , and the values of the parameters of the clustering algorithm were revealed.

The study of the distribution of computing resources by steps of the proposed algorithm was carried out by measuring the time costs at each step. The most resource-intensive is step 2, at which the clustering operation occurs in each layer. However, these operations are data-independent and can be performed simultaneously (in parallel) in all layers.

In some situations, due to the high density of points near the RS, a decrease in processing speed was observed. To solve this problem, an additional step was added to the algorithm to remove the central region of the point cloud with a radius of seven meters, corresponding to the space near the robot. Since existing methods are used to determine the permeability of a real RC in the indicated area, this operation does not affect the quality of the traffic control system and vehicle movement planning.

V. MACHINE VISION SYSTEM SOFTWARE

The software of the machine vision system is intended for returning and primary processing of images from installed sensitive elements vision system, calculating the type and location of found objects. Software input data are frames received from connected cameras. The frame is an image in the RGB (Bayer) format - a two-dimensional array of color filters.

The software output data is the information about the results of the object recognition according to the classifier. The software determines the object class, the distance to it and the angle. The output data is transmitted over UDP protocol.

System software provides:

- receiving frames from installed cameras;
- frame conversion;
- debugging cameras and calibration;
- objects detection and recognition on the received images;
- saving recognition results for further processing and analysis;
- calculating the location of recognized objects;
- transfer of information about recognized objects to the computing module.

The software can work without the hardware of the sensors, in the analysis mode of already recorded images. Setting parameters allows changing the number of cameras used and the level of image compression. The languages used for the image acquisition module are C++, Python. Open libraries Aravis, OpenCV, CUDA, Qt are used to implement software. The Qt library allows for networking, and also provides high performance for the system as a whole.

VI. EXPERIMENTAL RESULTS

The machine vision system digital tests were carried out in two directions:

- substitution of video images to identify objects;

- modeling the appearance of random objects and their approach to RS.

Fig. 3 and Fig. 4 show examples of screen forms of modeling various situations when obstacles are detected.

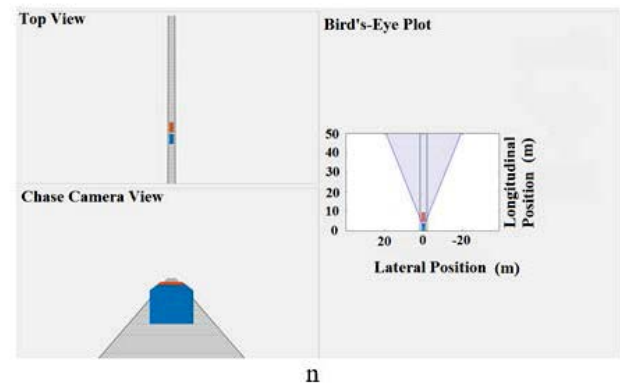


Fig. 3. A random object series along the RS route

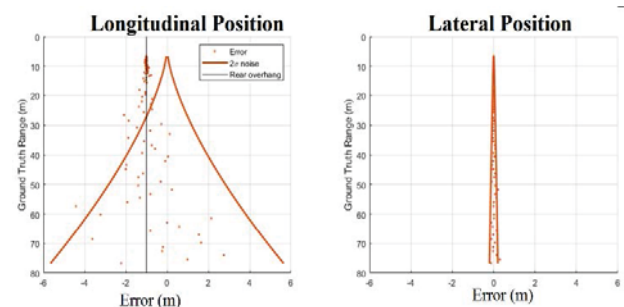


Fig. 4. Vision system response to a random object series

VII. CONCLUSION

The developed method of obstacle structural detection is implemented in a prototype robotic chassis as part of a common robotic system of agricultural vehicles on the KAMAZ platform.

The research development is supposed to be carried out in the direction of the intellectualization of the machine vision system using deep convolutional neural networks. Such an approach would provide a reliable classification of obstacles and generate the most efficient route robotic vehicle.

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